

Algebra II Complex Numbers and Polar Cheatsheet

Module 12

Complex Basics

$$i^2 = -1, \quad i^3 = -i, \quad i^4 = 1$$
$$\sqrt{-a} = i\sqrt{a}$$

Standard form:

$$a + bi$$

Conjugate:

$$a - bi$$

$$(a + bi)(a - bi) = a^2 + b^2$$

Complex Arithmetic

$$(a + bi) + (c + di) = (a + c) + (b + d)i$$

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i$$

Divide by multiplying by the conjugate of the denominator.

Polar Coordinates

$$x = r \cos \theta, \quad y = r \sin \theta$$
$$r^2 = x^2 + y^2, \quad \tan \theta = \frac{y}{x}$$

Choose θ using the quadrant of (x, y) .

Complex Polar Form

$$z = a + bi$$
$$r = \sqrt{a^2 + b^2}, \quad \tan \theta = \frac{b}{a}$$
$$z = r(\cos \theta + i \sin \theta) = r \operatorname{cis} \theta$$

Multiplication and Division in Polar Form

$$r_1 \operatorname{cis} \theta_1 \cdot r_2 \operatorname{cis} \theta_2 = r_1 r_2 \operatorname{cis}(\theta_1 + \theta_2)$$
$$\frac{r_1 \operatorname{cis} \theta_1}{r_2 \operatorname{cis} \theta_2} = \frac{r_1}{r_2} \operatorname{cis}(\theta_1 - \theta_2)$$

De Moivre

$$(r \operatorname{cis} \theta)^n = r^n \operatorname{cis}(n\theta)$$

Nth Roots

For

$$z = r \operatorname{cis} \theta$$

the n roots are:

$$z_k = r^{1/n} \operatorname{cis} \left(\frac{\theta + 2\pi k}{n} \right), \quad k = 0, 1, \dots, n-1$$

Quick Checks

- Complex conjugates are useful for division and real-coefficient polynomial roots.
- Polar angles are not unique.
- Negative r points in the opposite direction.
- Roots are evenly spaced around a circle.